CYCLIC SHIFT–JOIN DYNAMICS ON THE PARTITION LATTICE: MÖBIUS–BELL BASINS AND PRIMITIVE-CHORD DEPTH

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Abstract. Let rotation $\rho(i) = i + 1$ act on the partition lattice Π_n of $C_n = \mathbb{Z}/n\mathbb{Z}$, and iterate $J_n(\pi) = \pi \vee \rho\pi$. We analyze this finite functional graph. The t-th iterate is the join of the first $t+1$ translates of π, and its endpoint is the coset partition of the subgroup generated by all within-block differences. Hence the recurrent points are exactly the $\tau(n)$ cyclic block systems and $\zeta_{J_n}(z) = (1-z)^{-\tau(n)}$. If the endpoint subgroup has order $h \mid n$, its basin has the exact Möbius–Bell size

$$\sum_{d \mid h} \mu(h/d) B_d^{n/d}.$$

For $n \geq 3$, the sharp stabilization depth is $n - 2$. We also classify its deepest shell: it consists precisely of partitions with one two-element block $\{a, a+d\}$ satisfying $\gcd(d, n) = 1$ and all remaining blocks singleton, so its size is $n\varphi(n)/2$. The classification follows from a cut-defect argument, not enumeration. Exact controls make 1,916,206 assertions over all partitions through size ten and, independently, every binary cut through size twelve. Classical partition-lattice and permutation block-system results are owner-subtracted; external release remains on hold.

1. Introduction

A lattice automorphism and a join define a basic closure dynamics: start at x and repeatedly adjoin the next translate of the information already present. On a general finite lattice, this observation gives convergence but says little about endpoints, basins, or sharp time. The partition lattice over a regular cyclic action is rigid enough to determine all four.

Write Π_n for the partitions of $C_n = \mathbb{Z}/n\mathbb{Z}$, ordered by refinement, and let ρ be translation by one. We study the self-map

$$J_n(\pi) = \pi \vee \rho\pi.$$

The endpoint has a group-theoretic description: take every difference of two elements lying in one initial block, generate a subgroup, and partition C_n into its cosets. This description makes recurrence elementary and turns every basin into a divisor-lattice inversion of independent Bell choices. Sharp time requires a different argument. At the penultimate candidate time, a component cut can have only two untranslated defects. Parity on the cycles of a chord translation eliminates nonprimitive differences, while a primitive chord leaves a two-edge boundary that uniquely recovers that chord. This forces every deepest initial partition to be a single primitive-chord atom.

The resulting package has four concrete parts.

1. We give the exact iterate and the subgroup-coset endpoint of every state.
2. We classify recurrence, all iterate-fixed counts, and the finite Artin–Mazur zeta function.
3. We count every endpoint basin by Möbius inversion of Bell powers.
4. We prove the sharp depth $n - 2$ and classify all $n\varphi(n)/2$ deepest states.

Partition lattices, their joins, Bell enumeration, and ordinary Möbius inversion are classical [4]. In permutation-group theory, invariant partitions are block systems. Recent work studies

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lattices of invariant partitions [1], while Britnell and Wildon study when orbit partitions of group elements are closed under lattice operations [3]. Those structural facts including invariant closure by joined translates and the cyclic coset/subgroup correspondence, receive no credit. Residual scope is only the Möbius–Bell/depth conjunction. Internally this is neither P97's subset-sumset map nor P105's permutation-cycle pruning. A bounded search through 29 August 2026 did not locate the same package; search absence is not a novelty or priority certificate. External dissemination remains **HOLD**.

2. Exact iterates and subgroup endpoints

The join $\pi \vee \sigma$ is the least partition coarser than both π and σ. Equivalently, its blocks are the connected components of the graph that joins two vertices whenever they share a block in π or in σ. Relabelling by ρ is a lattice automorphism.

Definition 2.1 (shift–join map)**.** For $n \geq 1$ and $\pi \in \Pi_n$, define

$$J_n(\pi) := \pi \vee \rho\pi. \tag{1}$$

The stabilization depth of π is

$$\tau_n(\pi) := \min\{t \geq 0 : J_n^{t+1}(\pi) = J_n^t(\pi)\}.$$

Proposition 2.2 (consecutive-orbit join)**.** *For every $t \geq 0$,*

$$\boxed{J_n^t(\pi) = \bigvee_{j=0}^{t} \rho^j \pi.} \tag{2}$$

Proof. The formula holds at $t = 0$. Suppose it holds at time t. Because ρ preserves joins,

$$J_n^{t+1}(\pi) = J_n^t(\pi) \vee \rho J_n^t(\pi) = \left(\bigvee_{j=0}^{t} \rho^j \pi\right) \vee \left(\bigvee_{j=1}^{t+1} \rho^j \pi\right),$$

which is the join from $j = 0$ through $t + 1$. □

For a block B of π, take all differences $x - y$ with $x, y \in B$. Let

$$H(\pi) := \langle x - y : x, y \text{ lie in a common block of } \pi \rangle \leq C_n. \tag{3}$$

Write Q_H for the partition of C_n into cosets of a subgroup H.

Theorem 2.3 (endpoint, recurrence, and zeta)**.** *For every $\pi \in \Pi_n$, the stable endpoint is $Q_{H(\pi)}$. The fixed and recurrent states are exactly the coset partitions Q_H, one for each subgroup $H \leq C_n$. Consequently, for every $r \geq 1$,*

$$|\mathrm{Fix}(J_n^r)| = \tau(n), \tag{4}$$

and, as an identity in $\mathbb{Q}[[z]]$,

$$\boxed{\zeta_{J_n}(z) = \exp\left(\sum_{r \geq 1} \frac{|\mathrm{Fix}(J_n^r)|}{r} z^r\right) = (1 - z)^{-\tau(n)}.} \tag{5}$$

There are $\tau(n)$ one-cycles and no cycles of longer temporal period. Here $\tau(n)$ denotes the number of positive divisors of n.

Proof. The join of all n translates is translation invariant. It identifies 0 with every generator in (3), hence with all of $H(\pi)$, and translation identifies each x with $x + H(\pi)$. Conversely, every edge contributed by any translated block has difference in $H(\pi)$, so it cannot join distinct $H(\pi)$-cosets. The endpoint is therefore $Q_{H(\pi)}$.

A fixed partition satisfies $\rho\pi \leq \pi$. Relabelling preserves the number of blocks, so this refinement is equality; hence the partition is invariant under ρ and under the regular action of C_n. Its block containing 0 is a subgroup H: if $h \in H$, then translation by h sends the block containing 0 to the block containing h, which is the same block; this gives closure under addition, and finiteness gives inverses. All other blocks are its cosets. Conversely, each Q_H is translation invariant and fixed. The cyclic group has one subgroup of each order dividing n, so there are $\tau(n)$ fixed points.

Every update coarsens the current partition. A recurrent orbit in a finite partially ordered set can therefore contain only equal successive states, so recurrence equals fixedness. Thus every positive iterate has the same fixed set. Temporal Möbius inversion gives no cycles of length greater than one, and the defining series of Artin–Mazur [2] sums to (5). ☐

3. Exact Möbius–Bell basins

Let B_m be the m-th Bell number. For $h \mid n$, let

$$\mathcal{B}_n(h) := \{\pi \in \Pi_n : |H(\pi)| = h\}$$

be the basin of the unique fixed coset partition with block size h.

Theorem 3.1 (basin formula)**.** *For every $h \mid n$,*

$$\boxed{|\mathcal{B}_n(h)| = \sum_{d \mid h} \mu(h/d) B_d^{n/d}.} \tag{6}$$

In particular, the discrete partition has basin size one, and the one-block partition has basin

$$\sum_{d \mid n} \mu(n/d) B_d^{n/d}.$$

If $n = p$ is prime, the two basin sizes are 1 and $B_p - 1$.

Proof. Fix the subgroup of order h. The condition $H(\pi) \le H$ is equivalent to saying that every block of π lies inside one H-coset, or that π refines Q_H. There are n/h cosets, each of size h, and their restrictions can be partitioned independently. Hence

$$B_h^{n/h} = \sum_{d \mid h} |\mathcal{B}_n(d)|. \tag{7}$$

The subgroup lattice of a cyclic group is the divisor lattice. Ordinary Möbius inversion of (7) proves (6). The endpoint cases follow by substitution. ☐

The basin formula is not inferred from periodic data. Indeed, (5) sees only the number $\tau(n)$ of endpoints, whereas (6) retains the full divisor-indexed Bell profile.

4. Sharp depth and the primitive-chord shell

For a partition π, form a graph $G_t(\pi)$ on C_n: for every pair a, b in a common block of π and every $0 \le j \le t$, add the edge $\{a + j, b + j\}$. Its component partition is $J_n^t(\pi)$ by proposition 2.2.

Lemma 4.1 (one omitted translate is redundant)**.** *For every chord $e = \{a, a + d\}$, the graph formed by its first $n - 1$ translates has the same connected components as the graph formed by all n translates.*

Proof. Put $g = \gcd(d, n)$ and $\ell = n/g$. If $\ell \ge 3$, the full translate orbit of e is a disjoint union of g cycles of length ℓ. The first $n - 1$ translates omit one edge from one of these cycles, which does not change its vertex set or connectivity. If $\ell = 2$, each undirected edge occurs twice among the n translates; omitting one translate leaves every edge present. The case $d = 0$ contributes no chord. ☐

Applying the lemma to every within-block pair shows that $G_{n-2}(\pi)$ already has the terminal components. Thus $\tau_n(\pi) \le n - 2$ for $n \ge 3$. To determine equality, we need a cut lemma.

For a nonempty proper subset $S \subset C_n$, write $f = \mathbf{1}_S$. A chord $e = \{a, a + d\}$ is called *admissible for f* if

$$f(a + j) = f(a + d + j), \qquad 0 \le j \le n - 3. \tag{8}$$

Thus none of the first $n-2$ translates of e crosses the cut $S \mid (C_n \setminus S)$.

Lemma 4.2 (two-defect chord lemma)**.** *Assume $n \geq 3$ and $f : C_n \to \{0,1\}$ is nonconstant. Let $e = \{a, a+d\}$ be admissible for f.*

(1) *If $\gcd(d,n) > 1$, then $f(x) = f(x+d)$ for every $x \in C_n$.*
(2) *If $\gcd(d,n) = 1$, then the only two values of x for which $f(x) \neq f(x+d)$ are $a-2$ and $a-1$. The set $S = f^{-1}(1)$ has no nonzero translational stabilizer, and e is the unique primitive chord admissible for f.*

Proof. Set

$$D_d(f) := \{x \in C_n : f(x) \neq f(x+d)\}.$$

Admissibility gives $D_d(f) \subseteq \{a-2, a-1\}$. On every cycle of the permutation $x \mapsto x+d$, the number of changes of a binary function is even. If $\gcd(d,n) > 1$, the two possible defects lie in distinct cycles, because their difference 1 is not in the subgroup generated by d. Neither cycle can contain a single defect, so $D_d(f)$ is empty. This proves the first assertion.

Suppose now that d is primitive. Translation by d is one Hamilton cycle. The defect set cannot be empty because f is nonconstant, and its cardinality is even, so

$$D_d(f) = \{a-2, a-1\}. \tag{9}$$

Deleting these two boundary edges splits the Hamilton cycle into two paths; f is constant on each path and takes different values on them. Thus S is one nonempty proper cyclic interval in this Hamilton cycle. A nonzero rotation cannot preserve a single proper interval: a rotation preserving its two boundary edges either fixes both, hence is the identity, or swaps them and exchanges the interval with its complement. Therefore S has trivial translational stabilizer.

It remains to prove uniqueness. Let $s = |S|$, and let $q \in \{1, \ldots, n-1\}$ satisfy $qd \equiv 1 \pmod{n}$. The starting vertices $a-2$ and $a-1$ of the two boundary edges are separated by q steps in the d-cycle. The two components after deleting them have q and $n-q$ vertices. Hence $s \in \{q, n-q\}$ and

$$d \equiv \pm s^{-1} \pmod{n}. \tag{10}$$

Any other admissible primitive chord must satisfy the same equation, so it has the same undirected difference and the same Hamilton cycle. Its two omitted translates must be exactly the two cut edges determined by S. After orienting the difference as d, write those consecutive edges as $E_p = \{p, p+d\}$ and E_{p+1}. The initial chord must be E_{p+2}; reversing the orientation of d gives the same undirected edge. Thus the initial chord is uniquely e. □

Theorem 4.3 (sharp depth and complete deepest shell)**.** *For $n \geq 3$,*

$$\max_{\pi \in \Pi_n} \tau_n(\pi) = n-2. \tag{11}$$

Moreover, $\tau_n(\pi) = n-2$ if and only if π has one two-element block

$$\{a, a+d\}, \qquad \gcd(d,n) = 1, \tag{12}$$

and all other blocks are singleton. Consequently,

$$\boxed{\#\{\pi \in \Pi_n : \tau_n(\pi) = n-2\} = \frac{n\varphi(n)}{2}.} \tag{13}$$

For $n = 1, 2$, every partition is fixed and the maximum depth is zero.

Proof. The upper bound in (11) follows from lemma 4.1. If π is the atom in (12), its full translate graph is a Hamilton cycle. At time $n-3$, exactly two distinct edges of that cycle are absent, so the graph is disconnected. At time $n-2$, lemma 4.1 gives the full connected component. Hence this atom has depth $n-2$.

Conversely, suppose $\tau_n(\pi) = n-2$. Some component S of $G_{n-3}(\pi)$ is a nonempty proper subset of its component in the terminal translate graph. Let $f = \mathbf{1}_S$. Every chord joining two elements of one initial block is admissible for f, since all its first $n-2$ translates are edges of $G_{n-3}(\pi)$ and no such edge crosses a component cut.

If every initial chord had nonprimitive difference, the first part of lemma 4.2 would make f invariant under every within-block difference. It would then be invariant under the subgroup $H(\pi)$ they generate. Thus S would be a union of entire $H(\pi)$-cosets, contrary to its being a proper nonempty subset of one terminal coset. There is therefore an admissible primitive chord e. By the second part of lemma 4.2, S has trivial translational stabilizer and e is the only admissible primitive chord. A second nonprimitive chord would give a nonzero translational stabilizer by the first part of the lemma, while a second primitive chord would violate uniqueness. Hence e is the only chord in the equivalence graph of π. Since every block contributes all of its internal chords, π has exactly the one two-element block e and all remaining blocks are singleton.

Finally, choose $a \in C_n$ and a unit $d \in C_n$. This gives $n\varphi(n)$ oriented primitive chords. Each undirected chord is counted twice, by (a, d) and $(a + d, -d)$, proving (13). For $n = 1, 2$, direct inspection gives the stated endpoint convention. □

The proof of theorem 4.3 is independent of the Möbius–Bell basin route. Its only enumerative step is the final count of primitive chords; the if-and-only-if classification is a graph-cut theorem.

5. Recovery, controls, and limitations

The periodic ledger is deliberately incomplete as a parameter invariant: different values of n with the same divisor count have the same zeta function. Transient time restores recovery for $n \geq 3$, because $n = \max \tau_n + 2$. The two small cases have the same maximum depth, but their phase sizes $B_1 = 1$ and $B_2 = 2$ distinguish them.

Two analytic routes support the result. The *semilattice–subgroup route* proves the iterate, endpoint, recurrence, zeta, and basin formula. The *translated-graph route* realizes joins as components and proves sharp depth through chord cycles and binary cut defects. Neither route is a numerical interpolation of the other.

The deterministic control in `code/verify.py` supplies a third, finite check. It enumerates every restricted-growth string through $n = 10$, compares literal shift–join orbits with both a separately accumulated lattice join and an independently built translated graph, checks subgroup endpoints, all divisor-indexed basins, and the deepest classification state by state. It independently exhausts every binary cut through $n = 12$ for the defect lemma, the basin convolution through $n = 50$, and temporal Möbius/zeta identities through period 60. The fresh run contains 142,417 literal partitions and 1,916,206 exact integer assertions, with no randomness or floating-point calculation.

The results do not count every intermediate depth layer or every one-step fiber, and search absence does not establish priority. Residual scope is limited to the basin/depth conjunction proved here. Public release and specialist contact remain **HOLD**.

6. Conclusion

The map has exact subgroup endpoints and Möbius–Bell basins. Its deepest shell consists of primitive chords, has size $n\varphi(n)/2$, and occurs at sharp depth $n - 2$. Extending the basin and sharp-depth analysis to noncyclic regular actions is a natural next problem.

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